[INCH-POUND]
A-A-59389
3 August 1999
SUPERSEDING
MIL-G-80268A
25 August 1989

COMMERCIAL ITEM DESCRIPTION

# GRINDING MACHINE, TOOL AND CUTTER, FLOOR-MOUNTED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers floor-mounted, tool and cutter grinding machines. The machines are intended for use in grinding and sharpening conventional cutters, including milling cutters, reamers, counter bores, hobs, and other metal-cutting tools.

2. CLASSIFICATION. The grinding machines shall be of the following types, styles, and sizes. The type, style, and size to be furnished shall be as specified (see 7.2(b)).

Type I - Universal
Type II - Monaset

Style A - Manual workhead (type I only)
Style B - Powered workhead (types I and II)

Size - Sizes are shown in tables I and II

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3415

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes of type I grinding machines.

| Size[1] | Swing diameter, inches | Distance[2] (D) between centers, inches |
|---|---|---|
| 1 | 8 | 11 |
| 2 | 8 | 16 |
| 3 | 8 | 22 |
| 4 | 8 | 32 |
| 5 | 10 | 27 |
| 6 | 11 | 10 |
| 7 | 11 | 16 |
| 8 | 12 | 27 |
| 9 | 12 | 32 |

[1] Machine size and capacity shall not be less than the stated requirements of the size ordered nor greater than the next larger size and capacity shown.

[2]Tailstock centers set at the maximum distance apart (see figure 1 in paragraph 7.4.1).

TABLE II. Sizes of type II grinding machines.

| Size[1] | Swing diameter, inches | Distance[2] (D) between wheelhead spindle nose and workhead, inches |
|---|---|---|
| 1 | 9 | 4 |
| 2 | 9 | 10 |
| 3 | 12 | 6 |
| 4 | 13 | 37 |

[1] Machine size and capacity shall not be less than the stated requirements of the size ordered nor greater than the next larger size and capacity shown.

[2] Wheelhead spindle nose and workhead set at the maximum distance apart (see figure 2 in paragraph 7.4.2).

## 3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.1.1 Radii grinding capacities. Unless otherwise specified (see 7.2(c)), the type II grinding machine shall have convex and concave radii grinding capacities considered by the manufacturer as standard.

3.2 Components. The grinding machine shall include, as a minimum, the following components:



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3.2.1 Base. The base shall support other components of the grinding machine and incorporate materials, shape, and mass to minimize adverse vibration and deflection. The base of the grinding machine shall provide a means for leveling and anchoring the machine.

3.2.2 Machineways. Unless otherwise specified (see 7.2(d)), machineways for the worktable, cross-slide, column, and wheelhead shall be of the design considered by the manufacturer as standard for the machine. Unless otherwise specified (see 7.2(e)), machineways shall have protective covers and/or wipers to keep swarf and other debris from slide surfaces.

3.2.3 Wheelhead assembly.

3.2.3.1 Wheelhead assembly, type I. The wheelhead assembly shall consist of a spindle, spindle motor and drive mechanism, grinding wheel, support frame, and protective cowling. Unless otherwise specified (see 7.2(f)), the wheelhead motor horsepower shall be that considered by the manufacturer as standard for the machine. Unless otherwise specified (see 7.2(g)), the power transmission shall be a belt and step pulley system that provides for spindle speed changes. If required, a spindle that accommodates two grinding wheels shall be provided as specified (see 7.2(h)). Unless otherwise specified (see 7.2(i)), the wheelhead shall swivel 360 degrees in the horizontal plane and shall tilt a minimum of 15 degrees to the left and right of center in the vertical plane.

3.2.3.2 Wheelhead assembly, type II. The wheelhead assembly shall consist of a spindle motor and drive mechanism, spindle, grinding wheel, support frame, and protective cowling and shall be mounted on three sets of machineways for traverse in the longitudinal (X-axis), transverse (Y-axis), and vertical (Z-axis) directions. Unless otherwise specified (see 7.2(j)), the wheelhead motor horsepower shall be that considered by the manufacturer as standard for the machine. Unless otherwise specified (see 7.2(k)), the power transmission shall be a belt and step pulley system that provides for spindle speed changes.

3.2.3.3 Grinding wheels. Grinding wheels shall be supplied with the grinding machine as specified (see 7.2(l)). Unless otherwise specified (see 7.2(m)), grinding wheels shall conform to the shapes and sizes of American National Standards Institute (ANSI) B74.2, be marked in accordance with ANSI B74.13, and meet safety requirements of ANSI B7.1.

3.2.4 Worktable.

3.2.4.1 Worktable, type I. Unless otherwise specified (see 7.2(n)), the worktable shall have the length and width, the weight-holding capacity, and the longitudinal (X-axis) traverse considered by the manufacturer as standard. If required, the worktable shall have a minimum cross-slide (Y-axis) traverse as specified (see 7.2(o)). Unless otherwise specified (see 7.2(p)), the worktable shall swivel in the horizontal plane a minimum of 15 degrees either side of center. Unless otherwise specified (see 7.2(q)), the worktable shall have a single T-slot. When provided, the T-slot shall be the size considered by the manufacturer as standard and shall conform to the requirements of American Society of Mechanical Engineers (ASME) B5.1.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3.2.4.2 Worktable, type II. Unless otherwise specified (see 7.2(r)), the worktable shall have longitudinal (X-axis) traverse, cross-slide (Y-axis) traverse, and horizontal angular swivel considered by the manufacturer as standard. Unless otherwise specified (see 7.2(s)), the worktable shall have a single T-slot. When provided, the T-slot shall be the size considered by the manufacturer as standard and shall conform to the requirements of ASME B5.1.

3.2.5 Tailstocks, type I, style A. Unless otherwise specified (see 7.2(t)), the type I, style A grinding machine shall be equipped with tailstocks and centers considered by the manufacturer as standard. The maximum distance between tailstock centers shall be as shown in table I.

3.2.6 Workhead.

3.2.6.1 Workhead, type I, style A. If required, the type I, style A grinding machine shall have a manually operated workhead, which shall hold centers, chucks, and taper shank workpieces, as specified (see 7.2(u)). When a workhead is provided, the spindle taper shall be as specified (see 7.2(v)).

3.2.6.2 Workhead, type I, style B. The type I, style B grinding machine shall have a powered workhead, which shall hold centers, chucks, and taper shank workpieces. Unless otherwise specified (see 7.2(w)), the workhead shall have the drive motor horsepower, spindle speeds, spindle taper, and horizontal angular swivel considered by the manufacturer as standard.

3.2.6.3 Workhead, type II. The type II grinding machine shall have a powered workhead, which shall hold centers, chucks, and taper shank workpieces and be permanently mounted on the worktable. Unless otherwise specified (see 7.2(x)), the workhead shall have the drive motor horsepower, spindle speeds, spindle taper, vertical linear traverse, and vertical and horizontal angular swivel considered by the manufacturer as standard. The maximum distance between the wheelhead spindle nose and the workhead shall be as shown in table II.

3.2.7 Grinding fixtures, type I. If required, the type I grinding machine shall have one or more of the following grinding fixtures as specified (see 7.2(y)):

a. Tap grinding fixture.
b. Radial grinding fixture.
c. Spiral grinding fixture.
d. Universal work-holding fixture.
e. Tilting table fixture.

3.2.8 Stop dogs.

3.2.8.1 Stop dogs, type I. Unless otherwise specified (see 7.2(z)), two adjustable stop dogs per axis shall be provided for locating worktable linear axis motion reversal positions.

3.2.8.2 Stop dogs, type II. Unless otherwise specified (see 7.2(aa)), two adjustable stop dogs per axis shall be provided for locating workhead and wheelhead linear axis motion reversal positions.

3.2.8.3 Microswitches. When specified (see 7.2(bb)), electric microswitches shall be provided for accurate control of motion reversal positions.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3.2.9 Wheel dresser. Unless otherwise specified (see 7.2(cc)), a wheel dresser shall be provided and be mountable on the worktable for dressing the grinding wheel and removable after use.

3.2.10 Controls. Unless otherwise specified (see 7.2(dd)), the following manual controls and switches, as a minimum, shall be provided:

a. Manually operated hand wheels.
b. On-off power switches.
c. Forward-reverse switch.

3.2.10.1 Handwheels. Handwheel dials indicating machine linear position shall be graduated in increments no greater than 0.001 inch, have independent zero adjustment, and have continuous progression from the last graduation to the first graduation as the dial is rotated through the zero point. The graduations shall be permanently and legibly engraved or etched on a nonglare background.

3.2.10.2 Digital readout system. If required, a digital readout system shall be provided as specified (see 7.2(ee)). The system shall include position feedback devices and components necessary to provide a digital display of machine position over the working range of the X- and Y-axes. All components shall be protected from damage resulting from vibration, coolant splash, and grinding swarf or dust.

3.2.11 Lubrication system. Machine parts requiring periodic manual lubrication shall be readily accessible through doors, removable panels, or direct access. Unless otherwise specified (see 7.2(ff)), the lubrication system for the machineways shall be that considered by the manufacturer as standard for the machine.

3.2.12 Coolant system. If required, the grinding machine shall have a mist or flood coolant system as specified (see 7.2(gg)). The coolant system shall consist of a reservoir, pump, strainer, and all necessary piping. The system shall also be equipped with a method to control the rate of coolant flow from the reservoir and to direct coolant flow to the grinding area.

3.2.13 Electrical system. Unless otherwise specified (see 7.2(hh)), the electrical system shall conform to National Fire Protection Association (NFPA) 79. The machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(ii)). Unless otherwise specified (see 7.2(jj)), the control system power shall be provided by a separate 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.14 Electric motors. The grinding machine motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG 1. Motor controls shall conform to NEMA ICS 1 and NEMA ICS 3.

3.2.15 Optional accessories. Optional accessories shall be provided as specified (see 7.2(kk)).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3.3 Performance. Unless otherwise specified (see 7.2(ll)), the grinding machine shall meet the alignment requirements shown in table III when tested in accordance with sections 5.3 through 5.3.4.

TABLE III. Alignment requirements.

| Parameter | Tolerance |
|---|---|
| Spindle axial end play | 0.0002 inch |
| Spindle nose radial runout | 0.0002 inch TIR[1] |
| Tailstock spindles parallel to center of worktable T-slot | 0.0008 inch/foot, TIR |
| Worktable surface: | |
| Flatness | 0.0002 inch/foot, TIR |
| Parallel to transverse Y-axis of wheelhead spindle | 0.001 inch/foot, TIR |
| Parallel to longitudinal X-axis of wheelhead spindle | 0.001 inch/foot, TIR |
| Parallel to workhead and tailstock spindle centerlines | 0.001 inch/foot, TIR |
| Parallel to wheelhead spindle centerline over worktable swivel range | 0.001 inch/foot, TIR |
| Linear positioning (all linear axes) | ±0.0005 inch |
| Angular positioning (all rotary axes) | ±18 arc seconds |
| Cylindrical grind finish, Rt[2] (standard grinding wheel and mild steel) | 16 microinches |

[1] TIR is total indicator reading.
[2] Rt is the total roughness as defined in ASME B46.1.

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and weight shall not exceed the restrictions specified (see 7.2(mm)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.9. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI B11.9 and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(nn)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(oo)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(pp)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.
j. Serial number.

3.8 Lubrication plate or chart. When specified (see 7.2(qq)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

## 4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The grinding machine shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

## 5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The grinding machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(rr)), the grinding machine shall be tested in accordance with sections 5.3.1 through 5.3.4.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(ss)), the manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(tt)). Unless otherwise specified (see 7.2(uu)), the grinding machine shall be tested in the environmental conditions of the manufacturer's test facility.

5.3.2 Operational test. The grinding machine shall be operated in accordance with the manufacturer's standard operating procedures. The grinding machine shall be operated at no load for not less than 1 hour. Proper operation of all controls and features shall be verified during the trial period, after which the initial alignment test shall be conducted.

5.3.3 Alignment tests. Alignment tests shall be conducted on the grinding machine, and the results shall meet the requirements shown in table III. If optional tests are specified, the grinding machine alignment shall be rechecked upon completion of the optional tests, and the results shall meet the requirements shown in table III.

5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.2(vv)).

5.4 Acceptance. Unless otherwise specified (see 7.2(ww)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment tests shall be compared with the requirements shown in table III. Failure of the machine to meet the alignment requirements shown in table III shall be cause for rejection.

6. PACKAGING. Unless other wise specified (see 7.2(xx)), preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Sources of referenced documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

| | |
|---|---|
| ANSI B7.1 | Safety Requirements for the Use, Care, and Protection of Abrasive Wheels |
| ANSI B11.9 | Safety Requirements for the Construction, Care, and Use of Grinding Machines |
| ANSI B74.13 | Markings for Identifying Grinding Wheels and Other Bonded |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

Abrasives

ANSI B74.2 Specifications for Shapes and Sizes of Grinding Wheels and for Shapes, Sizes, and Identification of Mounted Wheels

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

American Society of Mechanical Engineers (ASME)

ASME B5.1 T-Slots, Their Bolts, Nuts, and Tongues

ASME B46.1 Classification and Designation of Surface Qualities

Applications for copies should be sent to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

National Electrical Manufacturers Association (NEMA)

NEMA MG 1 Motors and Generators

NEMA ICS 1 Industrial Control and Systems General Requirements

NEMA ICS 3 Industrial Control and Systems Factory Built Assemblies

Applications for copies should be sent to the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

National Fire Protection Association (NFPA)

NFPA 79 Electrical Standards for Industrial Machinery

Applications for copies should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

Occupational Safety and Health Association (OSHA)

OSHA 29 CFR PART 1910 Occupational Safety and Health Standards for General Industry

Applications for copies should be sent to the U. S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

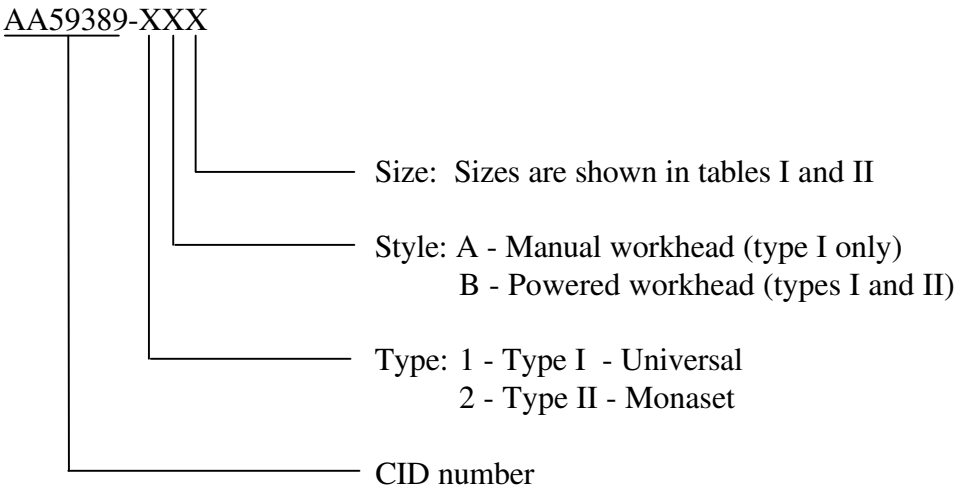
a. Title, number, and date of this document.
b. Type, style, and size of grinding machine, as specified (see 2).
c. Type II grinding machine convex and concave radii capacities, if different (see 3.1.1).
d. Machineways, if different (see 3.2.2).
e. Protective covers and/or wipers, if different (see 3.2.2).
f. Type I wheelhead motor horsepower, if different (see 3.2.3.1).
g. Type I wheelhead belt and step pulley system, if different (see 3.2.3.1).
h. Type I spindle with two grinding wheels, as specified (see 3.2.3.1).
i. Type I wheelhead swivel and tilt, if different (see 3.2.3.1).
j. Type II wheelhead motor horsepower, if different (3.2.3.2).
k. Type II wheelhead belt and step pulley system, if different (see 3.2.3.2).
l. Grinding wheels, as specified (see 3.2.3.3).
m. Grinding wheel shapes and sizes, marking, and safety requirements, if different (see 3.2.3.3).
n. Type I worktable length, width, weight-holding capacity, and longitudinal (X-axis) traverse, if different (see 3.2.4.1).
o. Type I worktable cross-slide (Y-axis) traverse, as specified (see 3.2.4.1).
p. Type I worktable swivel angle, if different (see 3.2.4.1).
q. Type I worktable T-slot, if different (see 3.2.4.1).
r. Type II worktable longitudinal (X-axis) and cross-slide (Y-axis) traverse and horizontal angular swivel, if different (see 3.2.4.2).
s. Type II worktable T-slot, if different (see 3.2.4.2).
t. Type I, style A tailstocks and centers, if different (see 3.2.5).
u. Type I, style A manual workhead, as specified (see 3.2.6.1).
v. Type I, style A workhead spindle taper, as specified (see 3.2.6.1).
w. Type I, style B powered workhead drive motor horsepower, spindle speeds, spindle taper, and horizontal angular swivel, if different (see 3.2.6.2).
x. Type II powered workhead drive motor horsepower, spindle speeds, spindle taper, vertical linear traverse, vertical and horizontal angular swivel, if different (see 3.2.6.3).
y. Type I grinding fixtures, as specified (see 3.2.7).
z. Type I stop dogs, if different (see 3.2.8.1).
aa. Type II stop dogs, if different (see 3.2.8.2).
bb. Microswitches, if required (see 3.2.8.3).
cc. Wheel dresser, if different (see 3.2.9).
dd. Manual controls, if different (see 3.2.10).
ee. Digital readout system, as specified (see 3.2.10.2).
ff. Lubrication system, if different (see 3.2.11).
gg. Coolant system, as specified (see 3.2.12).
hh. Electrical system industrial standard, if different (see 3.2.13).
ii. Electrical system voltage, as specified (see 3.2.13).
jj. Control system voltage, if different (see 3.2.13).
kk. Optional accessories, as specified (see 3.2.15).
ll. Performance, if different (see 3.3).
mm. Machine dimensions and weight, as specified (see 3.4).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

nn. System of units required (see 3.6).
oo. Dual system of units, if required (see 3.6).
pp. Nameplate, if different (see 3.7).
qq. Lubrication plate or chart, if required (see 3.8).
rr. Testing, if different (see 5.3).
ss. Test responsibility, if different (see 5.3.1).
tt. Certification of calibration, as specified (5.3.1).
uu. Test environmental conditions, if different (see 5.3.1).
vv. Optional tests, as specified (see 5.3.4).
ww. Acceptance, if different (see 5.4).
xx. Packaging requirements, if different (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.






Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

7.4 Examples of grinding machines.

7.4.1 Universal, type I, style A.

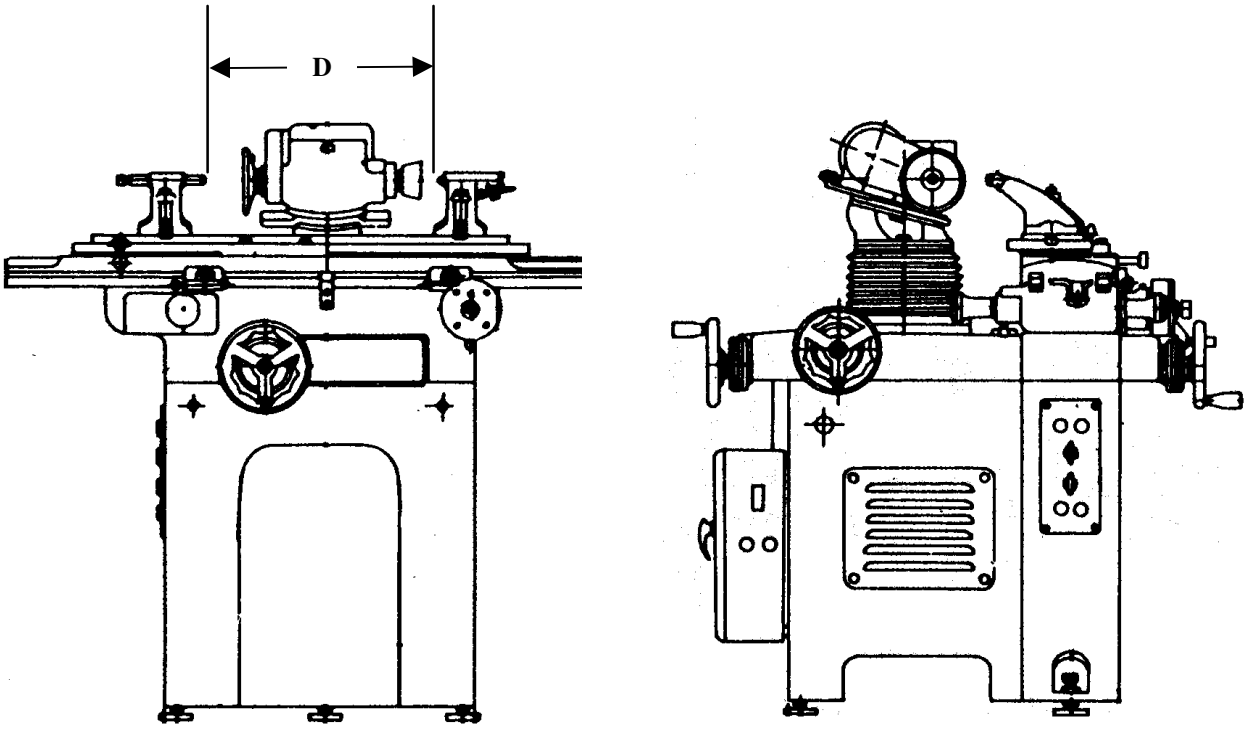



FIGURE 1. Typical manual universal tool and cutter grinding machine.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

7.4.2 Monaset, type II.

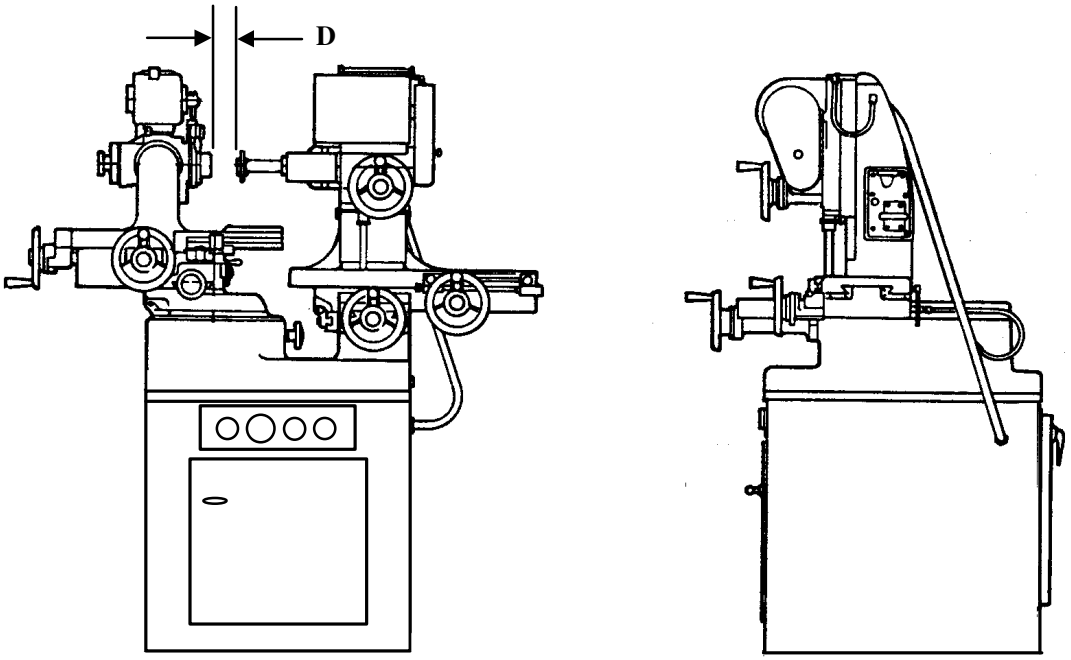



FIGURE 2. Typical monaset tool and cutter grinding machine.

MILITARY INTERESTS:

Custodians
Air Force - 99
Army - AL
Navy - SH

Reviewers
Navy - AS, MS, OS

PREPARING ACTIVITY:
DLA - GS

(Project 3415-0256)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.